

Marlow Rischmüller

APPLIED AI & MACHINE LEARNING - FOCUS ON NLP & LLMS

Bremen area, Germany

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M.Sc. graduate focused on the practical application of machine learning and LLMs. Experience fine-tuning open-source language models under real hardware constraints, alongside projects in NLP, computer vision and data analysis.

EDUCATION

M.Sc. Management Information Systems

10/2024 – 02/2026

University of Bremen

Final grade 1.96

B.Sc. Business Informatics

10/2019 – 05/2024

University of Bremen

Final grade 1.98

THESES

Master's Thesis

02/2025 – 08/2025

Specialising an Open Source LLM for Marine Engineering Applications within Hardware Constraints using LoRA Fine-Tuning

- Fine-tuned Microsoft Phi-3 on a domain dataset (10,000 entries, marine engineering & oceanography) via LoRA and supervised fine-tuning.
- Trained on an RTX 3090 Ti (24 GB); inference targeted under 8 GB to simulate consumer hardware.
- Improved METEOR from 0.300 to 0.348 on the domain benchmark with no degradation on the general MMLU benchmark - domain knowledge gained without catastrophic forgetting.
- Second strand: adapted the model to xVal number encoding; found LoRA fine-tuning insufficient and full pre-training necessary.

Python

PyTorch

HuggingFace Transformers

PEFT / LoRA

CUDA

Bachelor's Thesis

11/2023 – 02/2024

Simulation and Prediction of Saliency Maps for People with Impaired Visual Perception using Convolutional Neural Networks

- Adapted DeepGazeIIE to predict attention patterns under simulated impairments (reduced contrast, colour-vision deficiency, macular degeneration), using only the original image plus a modification flag.
- Trained on 10,000 images, evaluated on 4,012, scored with the SIM metric across single and combined modifications.
- Established that the approach does not meaningfully learn impairment-specific saliency, and analysed why - a deliberately reported negative result.

Python

PyTorch

OpenCV

torchvision

CUDA

FURTHER PROJECTS

Unified or Divided Through Sentiment? – A Party Analysis of Parliamentary Speeches

10/2023 – 02/2024

Sentiment and language analysis of 17,198 Bundestag speeches via a custom parsing pipeline, comparing intra-party unity. Project lead (team of 3), responsible for programming and data preparation.

Python

NLTK

pandas

seaborn

matplotlib

Replication: Is Accounting Useful for Forecasting GDP Growth?

10/2023 – 03/2024

Replicated and extended Datar et al. (2020): rebuilt the feature pipeline over 341 variables, validated with expanding walk-forward, and compared Elastic Net, Random Forest and a neural network. The Random Forest outperformed the original model at long horizons. Responsible for implementation and data preparation (team of 2).

Python scikit-learn pandas NumPy

OceanView - Beach Pollution Detection

10/2021 – 07/2022

Detection of beach litter on public webcam streams (no task-specific training). Sub-team coordination (5 of 11).

Python YOLO OpenCV Web Scraping

EXPERIENCE

Technical Operations & IT

2017 – 2024

Sun-Power, Delmenhorst - alongside studies

- Designed and implemented a distributed audio system: eight independent streams from one PC, individually selectable in up to 15 cabins.
- Modernised, extended and partly automated recurring statistical reporting.
- Systems maintenance and methodical fault diagnosis; onboarding and training of staff.

SKILLS

GENAI & LLMS

HuggingFace Transformers, PEFT / LoRA, supervised fine-tuning

DATA & ANALYSIS

pandas, NumPy, NLTK, matplotlib, seaborn

TOOLS

Git, Jupyter

SPOKEN LANGUAGES

German (native), English C1 (certified)

ML & DEEP LEARNING

PyTorch, scikit-learn, OpenCV, torchvision, CUDA

PROGRAMMING

Python (primary); SQL, Java (working knowledge)

FOCUS AREAS

NLP, LLM fine-tuning, computer vision, data analysis